Deepfake Identity Detection:
A Convolutional Neural Network Approach

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Contents

Introduction

- The rapid advancement of Generative Adversarial Networks (GANs), such as StyleGAN, has made it incredibly easy to generate hyper-realistic fake human faces.
- These AI-generated faces are actively being used to create fraudulent social media profiles, spread misinformation, and bypass basic digital identity checks.
- There is a critical need for systems that can quickly and accurately distinguish between genuine human photos and AI-generated deepfakes.

Objectives

Primary Objective

- Develop a web-based deepfake identity detector using CNN.

Secondary Objective

- To train a lightweight Convolutional Neural Network (CNN) capable of identifying spatial artifacts left by GANs in portrait images.
- To achieve a validation accuracy of over 90% on a large-scale dataset.
- To build a user-friendly web application with a "Strict-Mode" security gate to filter out non-human images before AI processing.
- To minimize false positives, ensuring real users are not incorrectly flagged as fake.

Literature Review

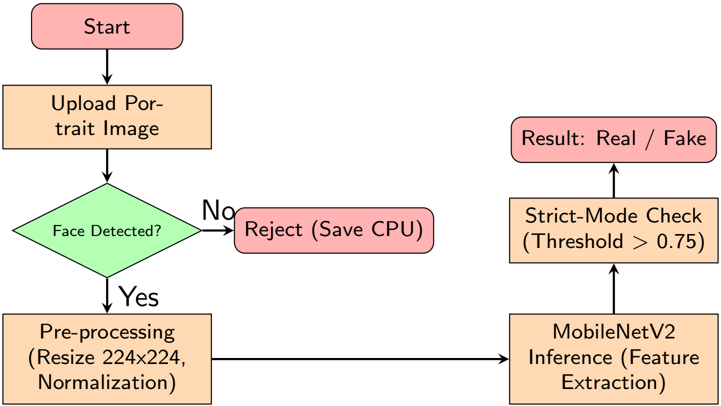
Title & Authors	Year	Key Contribution
MesoNet: A Compact Facial Video Forgery Detection Network Afchar et al.	2018	MesoNet is a small, low-parameter CNN detecting deepfakes with up to 98% accuracy. It stays fast by analyzing mesoscopic image properties.
Leveraging Edges and Optical Flow on Faces for Deepfake Detection Chintha et al.	2020	Using edge detection and optical flow achieved 99.31% accuracy. It proved that "cleaning" or improving data before it reaches the AI makes the system much more robust against low-quality or compressed images.
A Literature Review and Perspectives in Deepfakes Dagar et al.	2020	This survey emphasizes that future security depends on lightweight models that function independently of cloud infrastructure.

Proposed Methodology

The Pipeline:

1. **Input:** User uploads a portrait image.
2. **Face Validation Gate:** OpenCV (Haar Cascades) scans the image. If no human face is detected, the pipeline rejects the image to save compute power.
3. **Preprocessing:** Image is resized to $224 \times 224 \times 3$ and normalized (pixel values scaled to $0.0 - 1.0$).
4. **Inference:** The custom-tuned MobileNetV2 model analyzes the image for deepfake artifacts.
5. **Strict-Mode Output:** A threshold of > 0.75 is required to classify an image as "Real" to strictly prevent fake accounts from bypassing the system.

System Flowchart



Key Logic: The OpenCV gate acts as a "Computational Filter" before running the heavy CNN.

Proposed Model & Dataset

Dataset Details:

- Sourced from a balanced dataset of **140,000 images** (70k Real human faces, 70k StyleGAN-generated faces).

Model Architecture (Transfer Learning):

- **Base:** MobileNetV2 (Pre-trained on ImageNet, base layers frozen).
- **Custom Head:**
  - Global Average Pooling 2D (to flatten feature maps).
  - Dropout Layer at 0.5 (crucial for preventing overfitting).
  - Dense Output Layer with Sigmoid Activation (Outputs probability 0.0 to 1.0).

Web Application Technologies

Core Technologies:

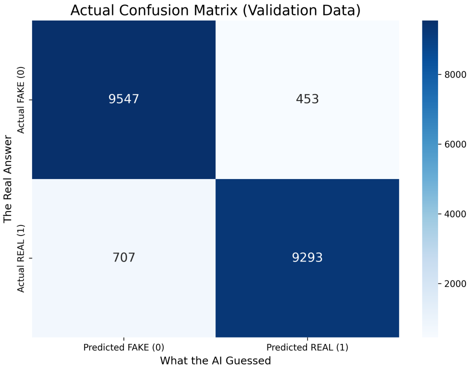
- **Frontend & UI:** Streamlit (Python)
- **Machine Learning:** TensorFlow / Keras
- **Computer Vision:** OpenCV, PIL (Pillow), NumPy

Application Modules:

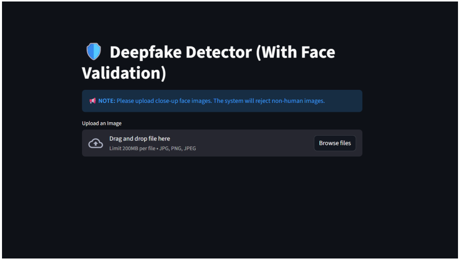
1. **Upload Module:** Handles JPG/PNG image ingestion.
2. **Security Gate Module:** OpenCV Haar Cascade implementation to block non-human objects (e.g., animals, landscapes).
3. **Prediction Module:** Loads the `.keras` weights and applies the 0.75 confidence threshold logic.

Experimental Results: Validation Metrics

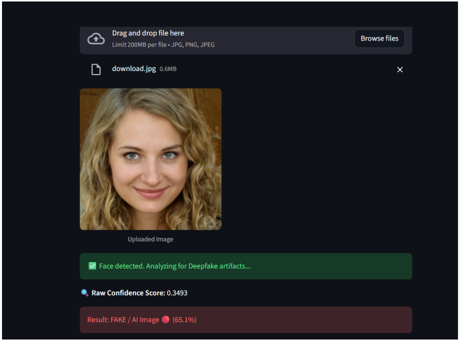
- Tested on 20,000 blind images.
- Correctly classified 9,547 Deepfakes and blocked and 9,293 Real humans verified. Misclassifications (707 false negatives and 453 false positives)
- **Final Accuracy: 94.2%**



1. The Upload Module



2. The Prediction Output



Conclusion

- Achieved an outstanding **94.2% accuracy** in detecting AI-generated faces.
- Proved **MobileNetV2** is the perfect lightweight model for fast, real-time scanning on standard devices.
- Boosted real-world reliability using an **OpenCV security gate** to filter out non-human images.
- Delivered a strong, lightning-fast defense against digital identity fraud.

Thank You!